



W13 · THE FOURTH ARABIC NLP CONFERENCE (ArabicNLP 2026) · EMNLP 2026 · BUDAPEST

CROSS-DIALECT ARABIC MACHINE TRANSLATION · ALEXANDRIAX-2026 SUBTASK 2

Capacity Discovery and Calibrated Preference Optimization for Cross-Dialect Arabic MT



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Rank #2 of 4 teams · spBLEU 25.84 / chrF++ 39.02 · best post-evaluation spBLEU 28.04

RESULTS AT A GLANCE

Second of four, and 0.36 from first after the deadline

25.84

Official spBLEU

rank 2 of 4 · chrF++ 39.02

28.04

Best post-evaluation

DPO Config B · 0.36 below 1st (28.40)

+8.9%

One rank + epoch step

beat three data-tuning rounds combined

70,139

Training pairs

3 trust tiers · 38 directions

Capacity looked more binding than data

Data rounds gave +3.8%, +2.7%, +1.8%; raising LoRA rank and epochs together gave +8.9%.

Planned training budgets were too short

Training past the scheduled epochs was consistently the largest lever.

DPO helped only with the right reward

Trusting every data tier hurt; a trust-restricted reward gave the best result.

THE TASK

MSA ↔ six dialects, with almost no parallel text

Each test instance is a (source dialect, sentence, target dialect) triple over short **banking-support** sentences. MSA appears only as a source or pivot, never as a scored target.

A blind test set of **3,574 rows** is scored with **spBLEU** (FLORES-200 SentencePiece) and **chrF++**.

The sentences are short and formulaic: the obstacle is uneven data scarcity, not translation difficulty in the usual sense.

Egyptian
no real or minable parallel text

Lebanese
no real or minable parallel text

Moroccan
mined pairs

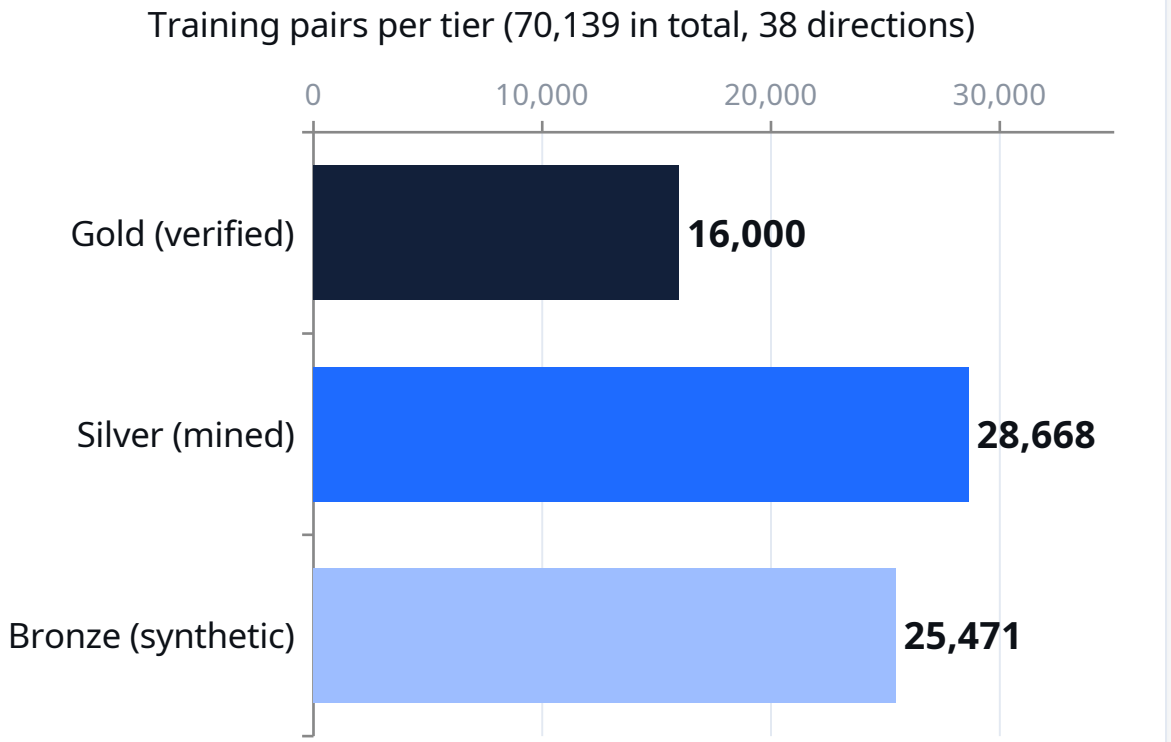
Palestinian
gold + mined pairs

Saudi
mined pairs

Tunisian
mined pairs

Egyptian + Lebanese ≈ 39% of test rows (our estimate from the test file, not an official breakdown). FLORES-200 has no one-to-one code for every dialect: Palestinian and Saudi use the closest broader Levantine and Peninsular codes.

A three-tier corpus built from ArBanking77



Gold · verified

Real, human-written MSA–Palestinian pairs: the only tier with a genuine ground-truth target.

Silver · topic-matched

Cross-dialect pairs mined by embedding similarity within a shared banking-intent bucket (MSA, Palestinian, Moroccan, Saudi, Tunisian): same topic, not a verified translation.

Bronze · back-translated

Synthetic Egyptian and Lebanese pairs, back-translated through the model itself and filtered for degenerate output.

Every experiment adapts facebook/nllb-200-distilled-600M: first with LoRA, later with full-parameter fine-tuning.

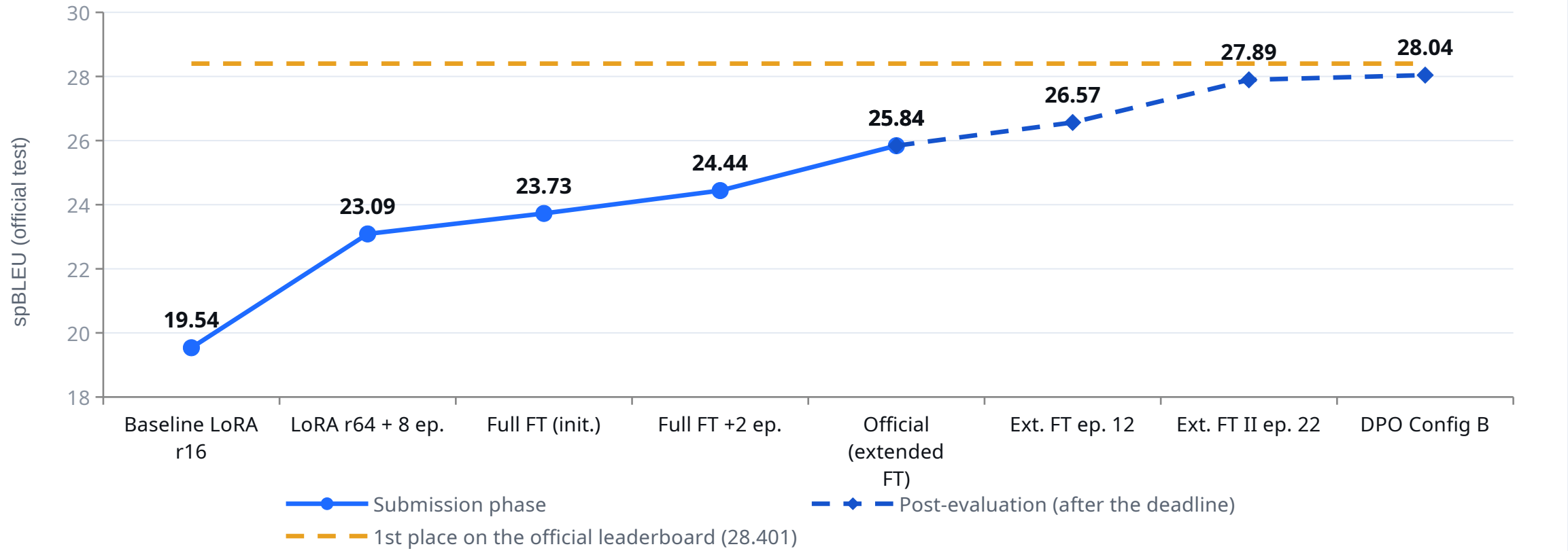
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01

Where the Gains Came From

Four phases, every configuration scored on the real leaderboard

Every configuration scored on Codabench



Official result 25.840 (chrF++ 39.019): 2.56 spBLEU behind 1st and 1.31 ahead of 3rd. Best post-evaluation checkpoint 28.042 (chrF++ 40.774) closes the gap to 1st to 0.36. Points are the exact milestones of the 24 real submissions (paper, Table 4).

Change the variable when returns flatten

P1

Data-recipe tuning (LoRA)

+3.8 → +2.7 → +1.8%

Widen silver/bronze caps, grow gold 2,500 → 16,000, fix the Tunisian artifact. Every round helped, each less than the one before.

P2

Capacity discovery

+8.9%

Data held fixed; LoRA rank 32 → 64 and epoch budget 5 → 8 raised together. Not ablated separately: it shows the data recipe had plateaued, not capacity alone.

P3

Full fine-tuning

trap found

Merge the best adapter, train all weights. Self-distilled bronze scored better offline and worse on the leaderboard: no dev set covers Egyptian or Lebanese.

P4

Training duration as a lever

+0.71 spBLEU

Validation loss was still falling when budgets expired: two extra epochs gave +0.71; open-ended rounds of up to 10 epochs gave the largest cumulative gain.

Each phase began because the previous one had stopped paying off.

02

Preference Optimization

Reward design decides whether DPO helps

Same recipe, three reward designs

Shared protocol: sample K = 4 candidates, add the training reference, score all, pair the best and worst as chosen / rejected. A and B start from the same Phase-4 checkpoint; C continues from B.

A — Uniform reward, β = 0.1

chrF++ to the reference + dialect-marker bonus for all six dialects. Trusts self-distilled bronze for Egyptian and Lebanese; a real reference is in the pool 99% of the time and usually wins, so DPO mostly re-teaches cross-entropy.

25.02 real spBLEU
regressed (then-best 27.89)

B — Trust-restricted reward, β = 0.3

No bronze references for the four dialects with gold or silver data. Egyptian and Lebanese get a reference-free reward: BGE-M3 similarity to the source + the marker bonus.

28.04 real spBLEU
best result overall

C — B repeated on its own output

Same reward, initialised from B, fresh candidates from the improved policy. Regressed on every offline metric tracked (silver_dev −5.6), so it was withheld.

— not submitted
offline regression

THE CENTRAL FINDING

Training dynamics did not predict real quality

Preference accuracy measures how well a policy satisfies its own reward, not whether that reward corresponds to real translation quality.

Config A

0.683
final-epoch preference acc.

25.02
real spBLEU regressed

Config B

0.691
final-epoch preference acc.

28.04
real spBLEU best overall

Config C

0.717
final-epoch preference acc.

—
real spBLEU offline regressed, not submitted

Reward auditing is not optional. And the objectives did not simply stack: continuing plain fine-tuning from B scored 27.51, below 28.04.

Bugs that raise no error

NLLB tokenizer

Called with src_lang, the fast tokenizer omits the FLORES-200 source-language token (otherwise it appends a spurious one). We build [lang_id, content, eos] by hand at training and inference.

Tunisian artifact

29.4% of raw Tunisian rows carried a stray leading underscore; the model reproduced it in 35% of its Tunisian predictions. Stripped before corpus construction.

peft merge-then-finetune

merge_and_unload() leaves merged weights frozen: the first full fine-tune trained nothing for two epochs (checkpoint byte-identical by checksum). Fix: requires_grad_(True) plus a trainable-parameter assertion.

bf16 DPO at learning rate 1e-5

Log-probability updates can fall below bf16's precision floor, silently making training a no-op. All DPO runs use fp32.

Clean negative results

MSA-pivot decoding

Lost to direct decoding in 11 of 12 non-MSA directions.

Diverse-beam MBR reranking

Improved 8 of 20 directions: noise level.

Cross-model consensus reranking

Failed: the weaker model was wrong more often, so trusting disagreement meant trusting it.

Decode hyperparameter sweep

Best alternative beat the default by ≈0.02 spBLEU.

Checkpoint averaging and model soups

Never beat the single best checkpoint; greedy soups reduced to their own input.

Exact-match retrieval

39% of test sources match a training source verbatim, but none from the gold tier: substitution risked harm.

Each was rejected by real leaderboard evidence, not by an offline proxy.

Key milestones of 24 real submissions

System iteration	spBLEU	chrF++
Baseline LoRA (r = 16)	19.536	35.010
Expanded LoRA (r = 64)	23.088	37.440
Full fine-tune (init.)	23.727	37.730
Full fine-tune (+2 epochs)	24.440	38.259
FT, extended (official submission)	**25.840**	**39.019**
FT, extended (ep. 12) †	26.565	39.610
FT, extended II (ep. 22) †	27.894	40.538
DPO Config A †	25.017	39.136
DPO Config B (best) †	**28.042**	**40.774**

† post-evaluation phase: scored on Codabench after the official deadline, not eligible for the ranking.

Setup

LoRA: r ∈ {16, 32, 64}, ≤ 8 epochs, lr 1e-4

Full FT: effective batch 32, lr 3e-5, bf16, early stopping (patience 2)

DPO: lr 1e-5, β ∈ {0.1, 0.3}, batch 6, 7 epochs, fp32

Decoding: beam 4, no-repeat 3-gram, repetition penalty 1.3

What to take away

Test capacity when data returns slow

When a data recipe's gains decelerate, test model capacity as an independent lever before assuming more data tuning is the answer.

Budget for longer training

Whenever validation loss had not plateaued, training past the scheduled epochs was the largest single lever.

Draw a trust boundary around the reward

Preference optimization for MT is only as good as the trust boundary around its reward's reference data; dynamics alone cannot reveal it.

FUTURE WORK & LIMITS

- Replace the lexicon + similarity proxy with a learned, reference-free quality-estimation reward.
- Test whether the capacity lever holds at a larger backbone.
- Offline dev metrics were directionally, not proportionally, reliable; Codabench reports aggregate scores only.

github.com/Momen-Ibrahim-Fawzy/ALEXIS-AlexandriaX-Subtask2-CrossDialectMT

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Thank You

Questions welcome

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PAPER
ACL Anthology

CODE
GitHub

aclanthology.org/2026.arabicnlp-sharedtasks.104
github.com/Momen-Ibrahim-Fawzy/ALEXIS-AlexandriaX-Subtask2-CrossDialectMT